

EL TORO MCAS, CA, USA

WMO: 722908

Lat: 33.667N

Lon: 117.733W

Elev: 117

StdP: 99.93

Time Zone: -8.00 (NAP)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.2	7.4	-12.0	1.4	15.2	-9.2	1.7	15.4	10.6	15.5	8.7	15.1	1.3	90	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.9	33.3	19.9	31.6	19.7	30.0	19.3	22.1	30.1	21.4	29.1	20.7	28.2	2.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.1	26.2	18.3	13.4	25.4	17.7	12.9	24.6	65.1	30.2	62.2	29.5	59.8	28.3	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.9	5.5	4.7	3.8	37.4	1.3	1.5	2.8	38.4	2.1	39.3	1.3	40.1	0.3	41.2
			0.2	24.2	1.4	1.3	-0.8	25.1	-1.6	25.9	-2.4	26.6	-3.4	27.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	14.3	14.6	15.4	17.2	18.5	20.5	22.5	23.5	22.8	20.3	17.2	14.1
	DBStd	4.15	2.53	2.84	2.51	2.67	2.29	2.29	1.80	2.16	2.28	2.88	3.01	2.34
	HDD10.0	4	1	2	0	0	0	0	0	0	0	0	0	2
	HDD18.3	617	129	111	97	52	26	6	0	0	0	9	55	132
	CDD10.0	3080	133	131	167	216	264	314	386	420	383	319	216	129
	CDD18.3	651	3	6	6	18	32	71	128	162	134	70	21	2
	CDH23.3	4618	41	62	74	198	174	440	759	1197	938	509	205	21
	CDH26.7	1473	4	11	15	44	44	132	194	431	343	199	54	2
Wind	WSAvg	1.7	1.8	1.9	1.8	1.8	1.8	1.8	1.6	1.6	1.4	1.6	1.7	1.9
Precipitation	PrecAvg	299	54	67	61	24	5	2	1	1	6	6	39	46
	PrecMax	650	191	345	260	95	41	8	9	21	44	43	170	200
	PrecMin	100	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	142	50	82	56	27	9	2	2	4	11	10	38	48
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.1	28.5	28.6	30.6	31.1	33.8	32.8	36.0	34.5	35.1	31.3	25.5
		MCWB	13.4	15.0	15.0	17.0	18.4	19.8	20.5	21.9	20.4	18.3	15.9	12.2
	2%	DB	24.2	25.2	25.3	27.7	27.3	30.1	30.5	33.0	32.4	31.5	27.8	23.2
		MCWB	11.9	13.7	14.6	15.7	17.8	19.6	20.9	21.1	20.4	17.7	14.4	12.4
	5%	DB	21.9	22.2	22.7	25.6	24.9	27.5	28.7	31.1	30.5	28.5	25.5	21.1
		MCWB	11.5	12.7	14.2	15.9	17.2	18.9	20.2	20.6	20.2	16.8	13.6	11.6
	10%	DB	19.8	19.7	20.5	23.2	23.3	25.6	27.3	29.1	28.6	25.8	23.4	19.1
		MCWB	10.9	12.6	13.7	15.2	16.3	18.1	19.7	20.4	20.0	16.9	12.9	11.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	16.6	17.1	19.0	19.8	21.4	22.6	23.7	22.8	21.2	17.6	15.6
		MCDB	18.9	24.2	24.4	26.9	28.3	30.2	30.6	32.0	30.3	31.0	26.4	20.7
	2%	WB	14.6	15.1	15.9	17.8	18.5	20.2	21.4	22.3	21.6	19.8	16.3	14.1
		MCDB	17.7	20.8	22.4	25.2	25.9	28.4	28.8	30.6	29.8	28.1	24.2	18.5
	5%	WB	13.9	14.3	15.1	16.8	17.6	19.3	20.7	21.6	20.8	18.6	15.4	13.3
		MCDB	17.5	19.1	20.8	23.5	24.0	26.8	27.4	29.2	29.0	25.4	22.5	18.1
	10%	WB	13.1	13.6	14.2	15.7	16.7	18.4	20.0	20.8	20.1	17.7	14.6	12.5
		MCDB	17.4	17.9	18.8	21.8	22.8	25.3	26.4	28.2	27.8	23.7	20.6	17.4
Mean Daily Temperature Range	5% DB	MDBR	9.6	9.6	9.7	10.5	9.0	9.8	9.8	10.9	10.8	10.7	10.9	9.9
		MCDBR	13.7	14.1	13.9	14.6	11.8	13.4	11.8	13.3	14.1	15.0	14.1	12.9
		MCWBR	7.5	7.5	7.5	7.6	6.3	6.7	5.5	6.2	7.1	8.0	8.0	7.6
	5% WB	MCDBR	9.7	10.4	11.0	12.2	10.2	12.2	10.6	11.5	12.3	12.0	11.4	9.9
		MCWBR	6.9	6.4	6.5	7.1	5.6	6.2	5.2	5.8	6.5	6.6	7.3	6.7
Clear-Sky Solar Irradiance	taub	0.313	0.321	0.345	0.354	0.353	0.367	0.376	0.362	0.346	0.344	0.305	0.303	
	taud	2.555	2.545	2.457	2.436	2.473	2.438	2.479	2.505	2.539	2.520	2.620	2.622	
	Ebn at Noon	906	934	932	933	931	913	903	914	918	894	911	900	
	Edh at Noon	81	90	106	114	111	115	109	105	97	91	75	72	
All-Sky Solar Radiation	RadAvg	3.21	4.09	5.47	6.63	7.10	7.51	7.47	7.11	6.01	4.56	3.55	2.91	
	RadStd	0.22	0.40	0.28	0.34	0.47	0.46	0.38	0.18	0.23	0.32	0.21	0.22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (49 neighbors)	+0.63	N/A	-1.70	+0.72	N/A	N/A	N/A	-112	+223	+112

Nomenclature: See separate page